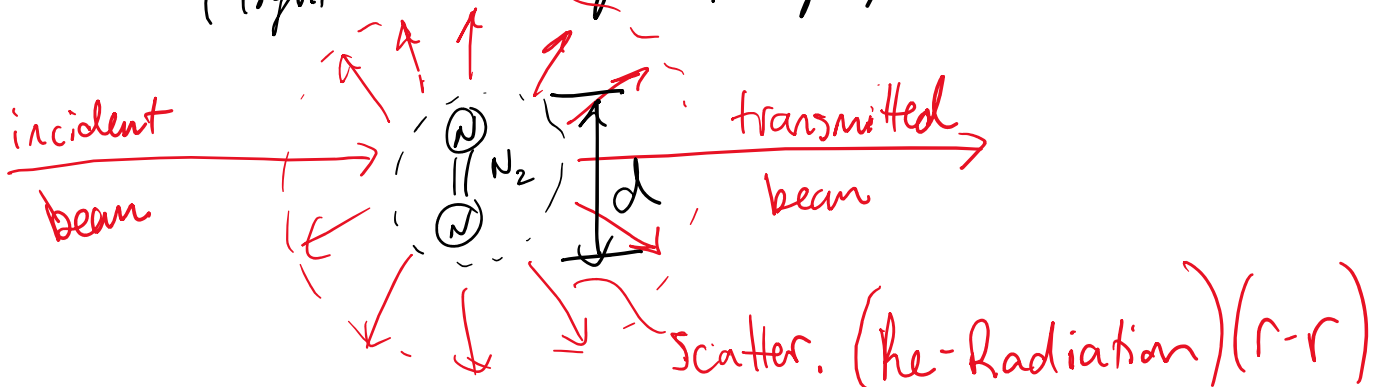


Scattering: a light beam is radiated away from the direction of propagation, due to interactions with "small" particles or defects in the medium.

2-types of Scattering:

1) Rayleigh Scattering: Particle size $\ll \lambda$ of light
(light interacting with gas)



$$\lambda_i \approx \lambda_{r-r}$$

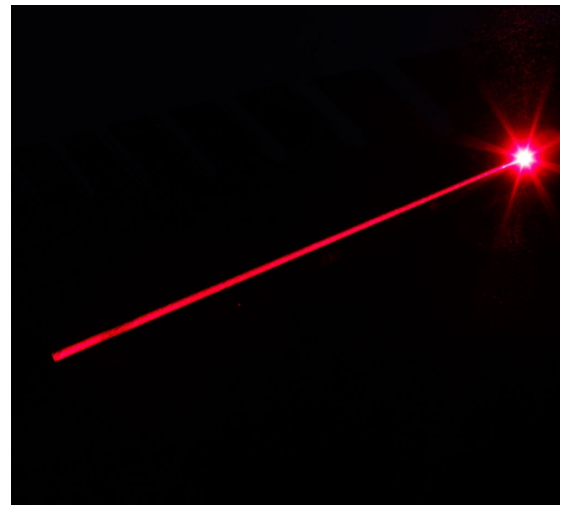
↑
incident wavelength

* Particle size $< \frac{1}{10} \lambda_i$
for Rayleigh Scattering

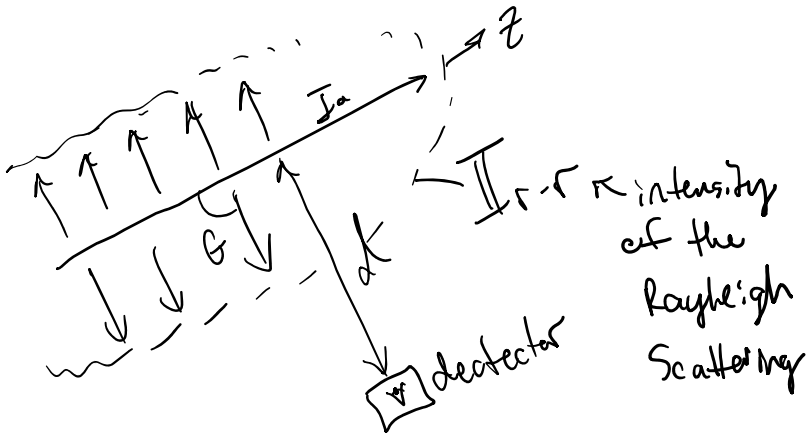
Rayleigh Scattering to see laser beam from the side



300mW Green Laser



3-5W Red Laser



$$I_{r-r} = I_0 \left[\frac{\delta(\pi)^4 N(\alpha_{r-r})(1 + \cos^2 \theta)}{\lambda^4 d^2} \right]$$

N = # of scattering interactions

α_{r-r} = Polarizability (how well the gas polarizes)

d = is the distance from beam to observer.

Polarizability proportional to

$$\alpha_{r-r}(\propto) N(a) \frac{1}{\lambda^4} \left(\frac{n^2 - n_0^2}{n^2 + n_0^2} \right)$$

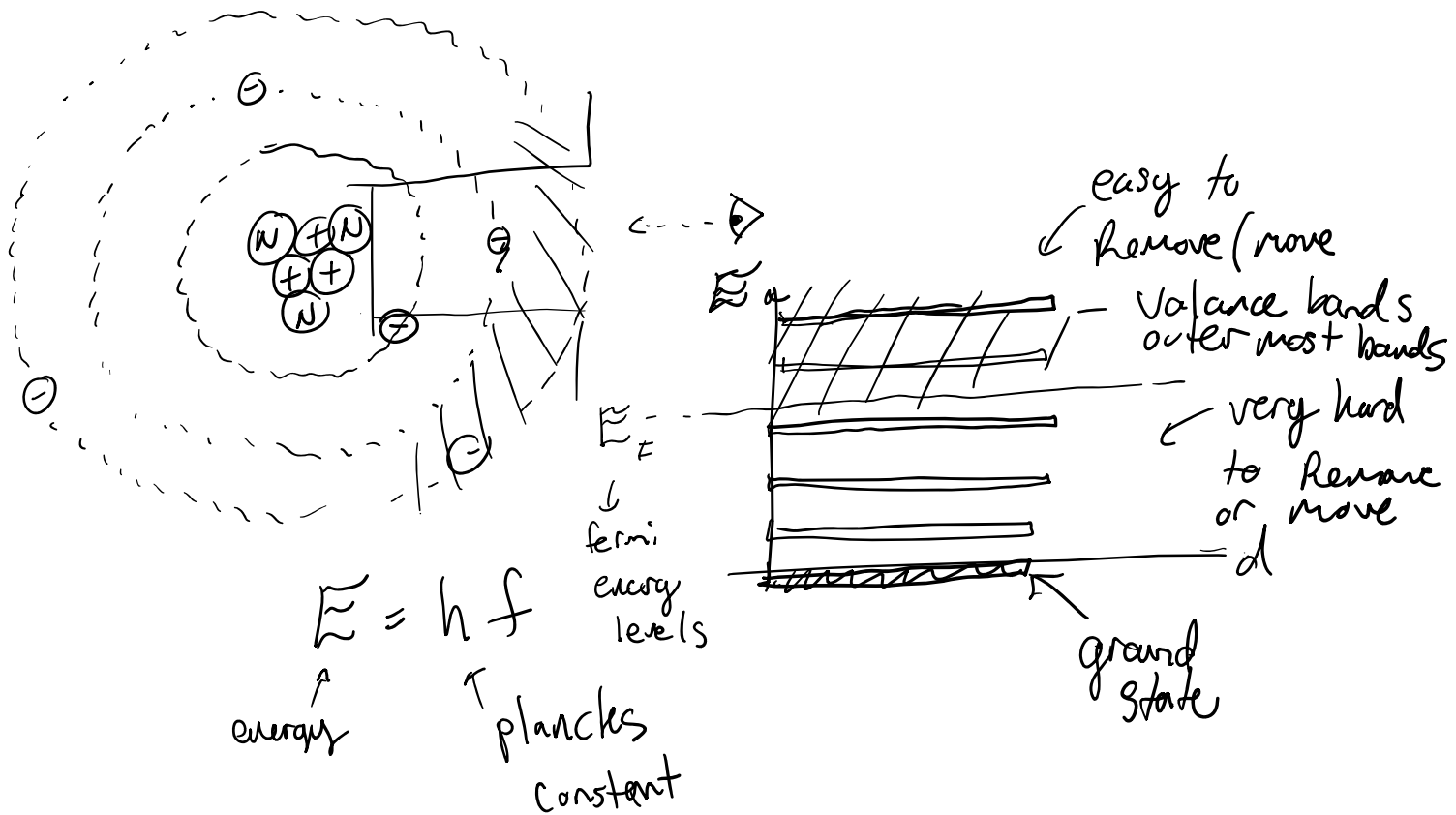
↑
radius of the particle

n is the index of refraction of the particle

n_0 is the index of refraction of the medium.

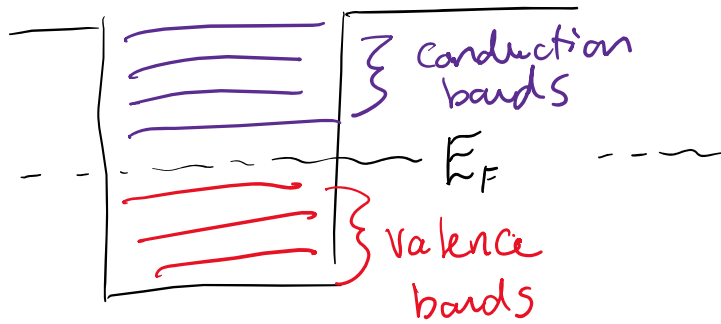
2.) Mie Scattering: occurs when the light
 $\lambda \geq$ the particle size
 (dust, fog, smoke, etc.)

Classic Model Atom

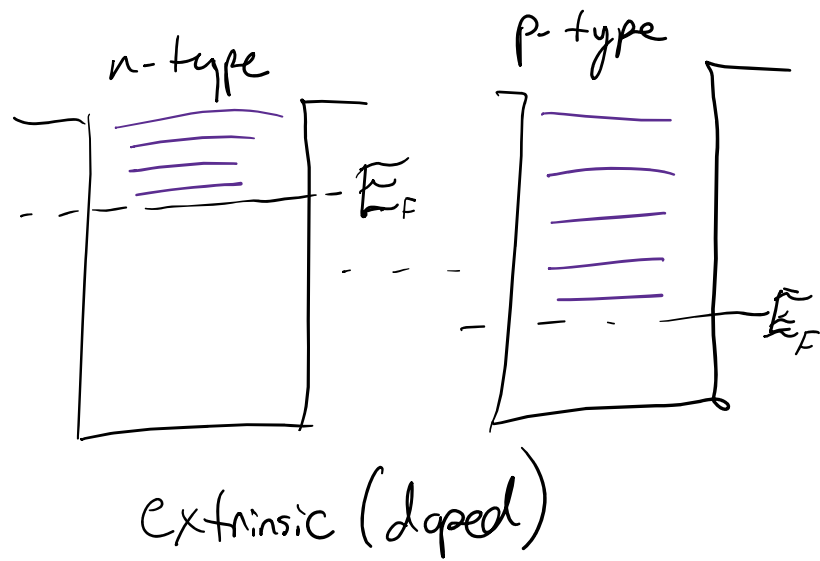


$$h = 6.626 \times 10^{-34} (\text{J} \cdot \text{sec})$$

P/N Junctions



Intrinsic Semi-conductors
(Non-doped)



Extrinsic (doped)

Extrinsic Semi-conductors

$$\text{conductivity } (\sigma_s) = q_e (N_d) \mu_e + q_e \frac{(n_i)^2}{N_d}$$

q_e = electron charge ($1.6 \times 10^{-19} \text{ C}$)

N_d = donor atom concentration

n_i = intrinsic concentration of electrons and holes in an undoped (pure) crystal

μ_e = is the electron drift mobility

μ_h = hole drift mobility

$$\mu_h \approx q_e N_d \mu_e$$

